



Critical Materials Recovery Capstone Project Requirements

Problem Statement

Organizations that receive surplus, obsolete, damaged, or unserviceable items often lack an efficient way to determine whether those items contain critical, strategic, precious, or otherwise valuable materials. Relevant information may be distributed across public item catalogs, technical descriptions, material references, lifecycle records, and commodity publications.

Manual research is slow and can produce inconsistent results. This creates missed opportunities to retain useful items, recover scarce materials, recycle valuable content, and document why a recommendation was made.

This capstone project will develop an open-source decision-support prototype that uses public and synthetic data to identify potentially valuable materials in cataloged items. The system will organize evidence in a knowledge graph, apply transparent classification rules, and present recommendations for human review. It will not treat an AI-generated inference as a confirmed material composition unless it is supported by a traceable source.

Project Objective

Build a reproducible prototype that can:

1. Accept a public item identifier or item description.
2. Retrieve matching public catalog information.
3. Connect the item to known or inferred material information.
4. Compare those materials with current critical-material references.
5. Recommend retention, reclamation, recycling, manual review, or ordinary disposal.
6. Show the evidence, rules, uncertainty, and source provenance behind the recommendation.

Functional Requirements

FR 1 Item Input

The system shall accept an item identifier, item name, or descriptive search phrase. Demonstration inputs shall contain only public or synthetic information.

FR 2 Public Catalog Search



The system shall search PUB LOG data or another approved public catalog source for matching item records.

FR 3 Data Normalization

The system shall normalize source fields into a common item model containing the item identifier, name, classification, characteristics, reference numbers, and applicable materialrelated indicators.

FR 4 Knowledge Graph

The system shall represent relationships among items, components, materials, material classifications, evidence sources, recovery methods, and recommendations.

FR 5 Material Evidence

The system shall associate an item with material information obtained from openly usable specifications, public materials datasets, or synthetic bills of materials.

Each material assertion shall include its source, retrieval date, license or reuse status, and evidence quality.

FR 6 Critical Material Comparison

The system shall compare identified materials with a versioned public list of critical or strategic materials, such as the current United States Geological Survey critical-minerals list.

FR 7 Item Classification

The system shall support the following recommendation categories:

- Retain for possible reuse
- Reclaim critical or strategic materials
- Recycle valuable materials
- Send for manual review
- Dispose through an ordinary approved process

FR 8 Explainable Results

Each recommendation shall display:

- The matched item record
- The materials identified
- The evidence supporting each material
- The classification rules that were applied
- The confidence or evidence-quality level



Missing or conflicting information

Whether human confirmation is required

FR 9 Human Review

The system shall require human review when material composition is inferred, evidence is incomplete, sources conflict, or a potentially hazardous material is identified.

A reviewer shall be able to accept, reject, or revise a recommendation and provide a reason.

FR 10 Recovery Estimate

Where suitable public or synthetic values exist, the system shall calculate an illustrative recovery estimate using:

Estimated material quantity

Expected recovery yield

Reference commodity value

Estimated processing cost

Estimated net recovery value

All estimates shall be labeled as illustrative and shall identify the date and source of each input value.

FR 11 Dashboard and Search Results

The system shall provide a simple interface showing item details, material findings, evidence, recommendation status, and aggregate counts by material and recommendation category.

FR 12 Audit Trail

The system shall record the input, source records consulted, rules applied, generated recommendation, human-review action, and timestamps for each analysis.

Audit records used in the capstone demonstration shall not contain personal information, credentials, internal network information, or real operational activity.

FR 13 Data Refresh and Versioning

The system shall record dataset versions and retrieval dates. It shall support refreshing public catalog and critical-minerals data without changing previously stored analysis results.

Data Requirements

Public Data Sources

Data source	Data used by the project	Usage requirement
PUB LOG Public Logistics Data	Non-restricted item identifiers, item names, supply classifications, characteristics, reference-number relationships, and permitted material-related indicators	Treat as publicly releasable data. Confirm current redistribution terms before including original bulk files in a public repository. Do not describe it as openly licensed unless an explicit license permits that use.
PUB LOG FLIS Data Electronic Reading Room	Downloadable data files, field descriptions, and record layouts	Record the download date and layout version because distributed fields may change.
USGS Critical Minerals List	Critical-mineral names, categories, and versioned criticality reference	Store the publication year and effective date. Do not assume that an older list remains current.
USGS Mineral Commodity Summaries	Mineral uses, production, supply-risk context, and annual market information	Use annual values as historical reference information, not as live prices.
NIST Materials Data Repository	Material properties and research datasets	Check the license attached to every selected record. Public availability does not give every record the same reuse license.
EPA RCRAInfo	Optional public information about regulated hazardouswaste handlers and compliance status	Verify bulk-download, API, and redistribution conditions before use.

Synthetic Data

The project shall create synthetic records for information that would normally come from operational systems. Synthetic records should be released under CC0 or CC BY 4.0 and must be clearly labeled as generated demonstration data.

Synthetic data shall include:

- Item condition and quantity
- Receipt, inspection, storage, and disposition events
- Generic location codes
- Example bills of materials

Material quantities and recovery yields
 Processing and transportation costs
 Review decisions and reviewer roles
 Sample photographs or generated item images
 Expected classifications for evaluation

Synthetic records shall not be generated by copying or slightly modifying restricted or proprietary records.

Core Data Entities

Entity	Required fields
Item	Public item ID, name, category, public characteristics, source, retrieval date
Component	Generic component name, parent item, public reference number, source
Material	Material name, standardized identifier, material category
Composition evidence	Item or component, material, percentage or quantity, evidence type, source, evidence quality
Criticality reference	Material, criticality category, publication, version, effective date
Lifecycle event	Synthetic item ID, event type, condition, quantity, timestamp, generic location code
Recovery method	Material, recovery process, expected yield, handling notes, source
Value estimate	Material quantity, reference value, recovery cost, calculation date, source
Recommendation	Item, category, explanation, confidence, human-review requirement
Review decision	Recommendation, generic reviewer role, decision, reason, timestamp
Provenance record	Source URL or record ID, publisher, retrieval date, license or reuse status
Audit event	Analysis ID, action, rules executed, timestamp, result

Data Quality Requirements

The system shall:

- Preserve the original source identifier for every imported public record.
- Distinguish confirmed facts from inferred values and synthetic values.
- Reject records that lack required provenance.
- Detect missing, duplicated, malformed, or conflicting identifiers.
- Record units for all quantities, percentages, prices, distances, and weights.
- Normalize material names while preserving the original source value.
- Prevent an inference from overwriting a confirmed source value.
- Display the age and version of data used in each recommendation.

Data Protection Requirements

The project repository and demonstration data shall not contain:

- Personal names, signatures, email addresses, telephone numbers, or personnel records
- Credentials, tokens, passwords, certificates, or access-control information
- Internal IP addresses, hostnames, system logs, or security events
- Real facility locations, stock levels, movements, or operational activity
- Controlled, export-restricted, service-unique, or proprietary technical data
- Non-public bills of materials, technical data sheets, maintenance records, or supplier records
- Images containing people, badges, screens, location clues, labels, or embedded metadata

Public supplier or organization records shall be minimized to the fields required for item matching. Personal contact details and residential addresses shall not be imported.

Nonfunctional Requirements

NFR 1 Open-Source Implementation

The application source code shall use an open-source license selected by the student team. Third-party libraries shall have licenses compatible with the selected project license.

NFR 2 Reproducibility

The repository shall include setup instructions, dataset acquisition instructions, synthetic-data generation instructions, and a repeatable evaluation procedure.

NFR 3 Explainability



Users shall be able to trace every recommendation to its rules and evidence without relying on hidden model reasoning.

NFR 4 Privacy and Security

The system shall use least-privilege access, validate uploaded files, sanitize logs, and avoid storing secrets in source control.

NFR 5 Usability

A student evaluator shall be able to search for an item, review its evidence, and understand the recommendation without specialist training.

NFR 6 Maintainability

Data ingestion, classification rules, graph storage, recommendation logic, and the user interface shall be separated into testable components.

Evaluation Requirements

The capstone evaluation dataset shall contain at least 100 public or synthetic item records with documented expected outcomes.

The final demonstration shall report:

- Catalog-match success rate

- Material-identification precision and recall where labels are available

- Percentage of recommendations with complete provenance

- Percentage of uncertain cases correctly sent for human review

- Reviewer agreement with system recommendations

- Average processing time per item

- Known data limitations and error cases

PUB LOG Classification

PUB LOG is a free source of publicly releasable logistics information. It is appropriate for a student prototype using non-restricted catalog fields. The project documentation should call it **public data** or **publicly releasable data**, not automatically **open-source data**.

Before redistributing PUB LOG files or a substantial derived dataset, the team shall review the current publisher terms. Application and security log files are unrelated to PUB LOG and shall not be treated as public data.